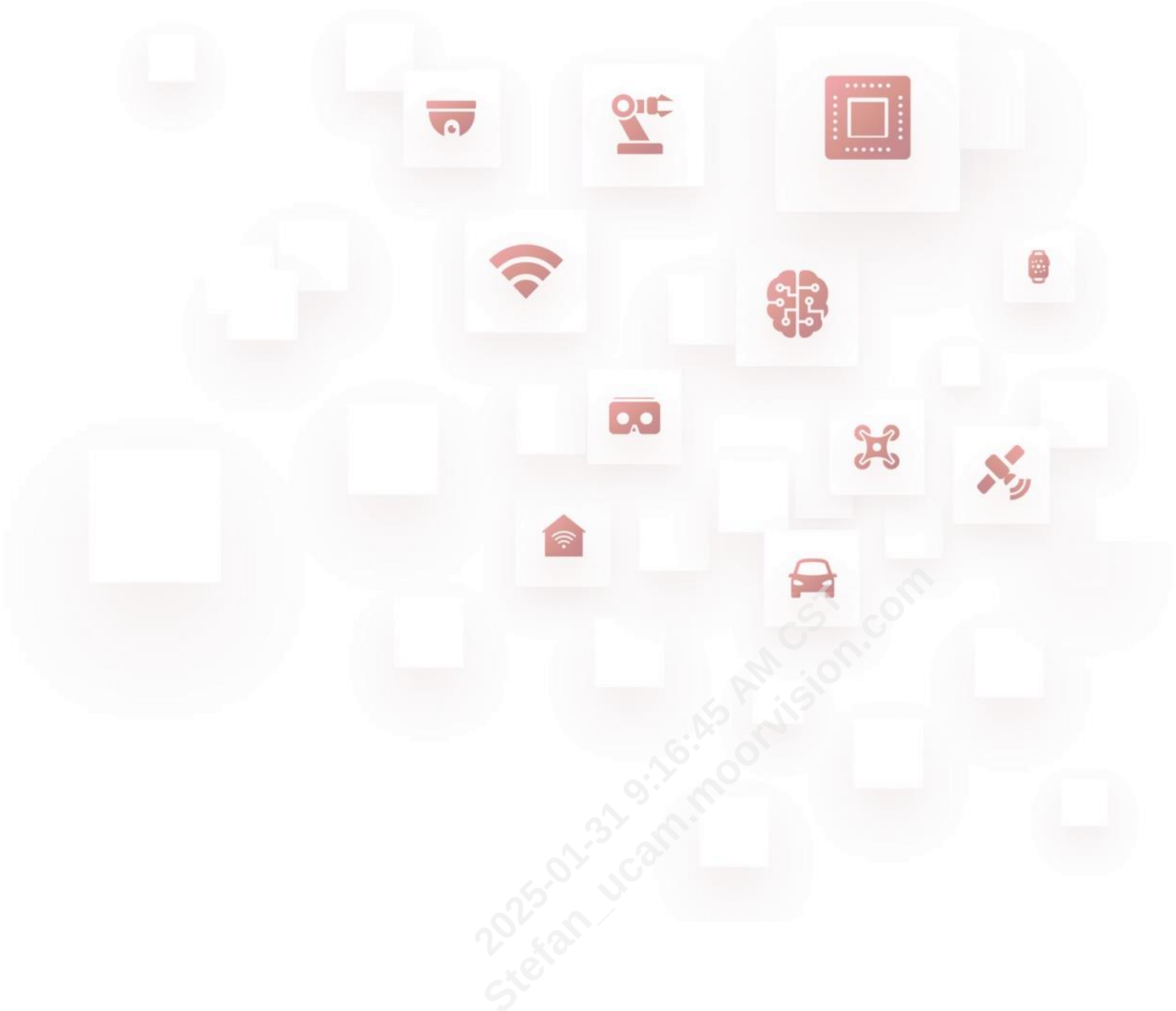
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Thundercomm TurboX C610 Open Kit Linux Embedded

Application SDK User Manual

Rev. V1.0
Oct 29, 2021

DN: tc-g2le-23110

Revision History

Revision	Date	Description
1.0	Oct 29, 2021	Initial release.

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- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of your computer, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document is an Application SDK instruction manual for TurboX C610 Open Kit 64-bit system. It describes how to download and install Application SDK, as well as how to develop an application in the Application SDK.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

This Application SDK is based on FC2.r001004 release and later versions. For more information, refer to *TurboX C410 C610 OK LE Software Release Notes_(for FC2.r001004)_20210402_V1.2*.

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Chapter 2. Compile and Flash 64-bit System

Before installing the new version of 64-bit Application SDK that supports OpenCV, you need to flash the 64-bit system to the C610 board first.

Step 1. Compile 64-bit System.

1) Confirm the system image compilation script.

```
$ cd turbox/tools/build_script/platform
```

2) Confirm the corresponding script file.

```
$ git diff .
diff --git a/build_hlos.sh b/build_hlos.sh
index 755380d..d932924 100755
--- a/build_hlos.sh
+++ b/build_hlos.sh
@@ -64,7 +64,7 @@ USAGE
 OPTIONS=""
 #TARGET_PRODUCT="msm8996"

-PRODUCT="qcs610-odk"
+PRODUCT="qcs610-odk-64"
 VARIANT="debug"

 SRC_PATH="apps_proc"
diff --git a/zip_fastboot_build.sh b/zip_fastboot_build.sh
index cf0b7be..a1cae2f 100755
--- a/zip_fastboot_build.sh
+++ b/zip_fastboot_build.sh
@@ -55,7 +55,7 @@ Example:
 USAGE
 }

-PRODUCT=qcs610-odk
+PRODUCT=qcs610-odk-64
 VARIANT=debug

 TARGETDIR=$meta_dir/turbox/output
diff --git a/zip_flat_build.sh b/zip_flat_build.sh
index 009aa1d..573c44c 100755
--- a/zip_flat_build.sh
+++ b/zip_flat_build.sh
@@ -55,7 +55,7 @@ Example:
 USAGE
 }

-PRODUCT=qcs610-odk
+PRODUCT=qcs610-odk-64
 VARIANT=debug

 TARGETDIR=$meta_dir/turbox/output
diff --git a/zip_meta_build.sh b/zip_meta_build.sh
index fed834e..4e85b5a 100755
--- a/zip_meta_build.sh
+++ b/zip_meta_build.sh
@@ -55,7 +55,7 @@ Example:
 USAGE
 }

-PRODUCT=qcs610-odk
+PRODUCT=qcs610-odk-64
 VARIANT=debug

 TARGETDIR=$meta_dir/turbox/output
```

3) Compile the 64-bit system with the compilation script.

```
{your_C610_root_dir}$ ./turbox_build.sh -v debug -a -l
```

Step 2.Flash the 64-bit system to the C610 board.

1) Confirm the image flashing script.

```
{your_C610_root_dir}$ vim turbox_flash.sh
diff --git a/turbox_flash.sh b/turbox_flash.sh
index c5cccc0..35399b1 100755
--- a/turbox_flash.sh
+++ b/turbox_flash.sh
@@ -140,9 +140,9 @@ flash_userdata() {
}
flash_hlos() {
-   fastboot   flash   abl_a   apps_proc/build-qti-distro-fullstack-${VARIANT}/tmp-
glibc/deploy/images/qcs610-odk/qti-multimedia-image/abl.elf
-   fastboot   flash   boot_a   apps_proc/build-qti-distro-fullstack-${VARIANT}/tmp-
glibc/deploy/images/qcs610-odk/qti-multimedia-image/boot.img
-   fastboot   flash   system_a   apps_proc/build-qti-distro-fullstack-${VARIANT}/tmp-
glibc/deploy/images/qcs610-odk/qti-multimedia-image/system.img
+   fastboot   flash   abl_a   apps_proc/build-qti-distro-fullstack-${VARIANT}/tmp-
glibc/deploy/images/qcs610-odk-64/qti-multimedia-image/abl.elf
+   fastboot   flash   boot_a   apps_proc/build-qti-distro-fullstack-${VARIANT}/tmp-
glibc/deploy/images/qcs610-odk-64/qti-multimedia-image/boot.img
+   fastboot   flash   system_a   apps_proc/build-qti-distro-fullstack-${VARIANT}/tmp-
glibc/deploy/images/qcs610-odk-64/qti-multimedia-image/system.img
}
```

2) Flash the 64-bit system image to the C610 board.

```
$ adb reboot bootloader
$ ./turbox_flash.sh -a
$ fastboot reboot
```

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Chapter 3. Set up the Development Environment

Step 1. Download the Application SDK.

Step 2. Unpack the Application SDK.

```
$ tar -xzvf Turbox-C610_Application-SDK_v1.0.tar.gz
$ cd Turbox-C610_Application-SDK_v1.0
$ ls
oecore-x86_64-aarch64-toolchain-nodistro.0.host.manifest
oecore-x86_64-aarch64-toolchain-nodistro.0.target.manifest
oecore-x86_64-aarch64-toolchain-nodistro.0.sh
oecore-x86_64-aarch64-toolchain-nodistro.0.testdata.json
```

Step 3. Execute the following command.

```
$ ./oecore-x86_64-aarch64-toolchain-nodistro.0.sh
```

Step 4. To choose the default target directory, press **Enter** and input **Y**.

```
QTI Linux full reference stack distro targeting performance builds. SDK installer
version nodistro.0
==========================================================
Enter target directory for SDK (default: /usr/local/oecore-x86_64):
You are about to install the SDK to "/usr/local/oecore-x86_64". Proceed[Y/n]? Y
[sudo] password for <host>: (input your sudo password)
Extracting
SDK.....................................................................................
.....................................................done
Setting it up...done
SDK has been successfully set up and is ready to be used.
Each time you wish to use the SDK in a new shell session, you need to source the
environment setup script e.g.
$ . /usr/local/oecore-x86_64/environment-setup-aarch64-oe-linux
$ . /usr/local/oecore-x86_64/environment-setup-armv7ahf-neon-oe-linux-gnueabi
```

Step 5. Configure the system environment:

```
$ source /usr/local/oecore-x86_64/environment-setup-aarch64-oe-linux
```

Step 6. Compile the command.

```
$ $CC test.c -o test
```

➲ **NOTE**:

- The *test.c* file is provided by the customer.
- To compile the OpenCV program, execute the following command:

```
$ $CC opencv-test.cpp -xc++ -lstdc++ -shared-libgcc -lopencv_core -
lopencv_imgcodecs -lopencv_imgproc -o opencv-test
```

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Chapter 4. Develop the Application

After installing the Application SDK, you can use it to build your own C/C++ programs.

Proceed with the steps below to build a sample C++ program with OpenCV and execute it on the board.

Step 1. Prepare a C++ program file with the following content. The commands below will make a picture gray with the OpenCV library.

```
#include <opencv2/opencv.hpp>
int main()
{
        cv::Mat image, image_gray;
        image = cv::imread("lenna.png", cv::IMREAD_COLOR);
        cv::cvtColor(image, image_gray, cv::COLOR_BGR2GRAY);
        cv::imwrite("lenna_gray.png", image_gray);
        return 0;
}
```

Step 2. Build the application.

1) Set up the environment.

```
$ source /usr/local/oecore-x86_64/environment-setup-aarch64-oe-linux
```

2) Build a hello application with `$cc` command.

```
$ $CC opencv-test.cpp -xc++ -lstdc++ -shared-libgcc -lopencv_core -
lopencv_imgcodecs -lopencv_imgproc -o opencv-test
```

➲ **NOTE**: For C++ programs, please use `$cxx` command.

Step 3. Transfer the application to the board.

1) Connect your PC to the board via micro-USB and ensure that the adb port is available.

2) Remount the root directory as writable.

```
$ adb root
$ adb disable-verity
$ adb reboot
$ adb root
$ adb remount
$ adb shell mount -o remount,rw /data
```

3) Transfer the hello application from PC to the board with adb command.

```
$ adb push opencv-test /data/
$ adb push lenna.png /data/
```

Step 4. Execute the *OpenCV-test* application in adb shell environment.

```
$ adb shell
# cd data
# chmod +x opencv-test
# ls
audio         data_qcmap    misc          persist       vendor
coredump      lenna.png     opencv-test   tombstones    www
# ./opencv-test
# ls
audio           lenna.png       opencv-test     vendor
coredump        lenna_gray.png  persist         www
data_qcmap      misc            tombstones
```

Step 5. Then the *OpenCV-test* program generates another picture. You can pull it out with the following command.

```
$ adb pull /data/lenna-gray.png
```

Step 6.Refer to the comparison below of the pictures before and after the sample program is executed.

Before



After



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Appendix 1. Notices

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